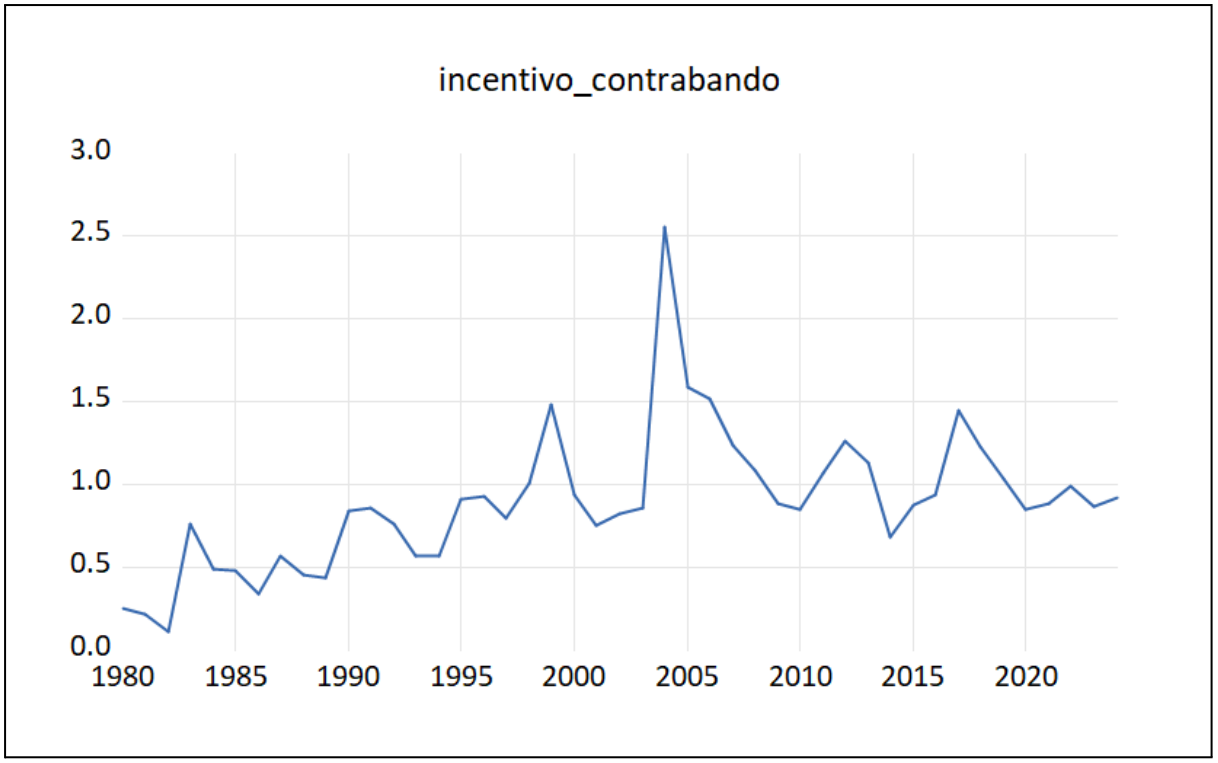



Se nota una leve tendencia entre 1980 y principios de los 2000 que termina con la presencia de un salto abrupto en el año 2004 que puede deberse a la guerra civil ocurrida en Costa de Marfil, que ocurrió entre los años 2002-2007, pero que tuvo su apogeo en el 2004. A partir del 2010 se observa cierta estabilidad.

Analizo los test de Dickey Fuller para verificar la existencia de raíz unitaria, empezando con el modelo con tendencia determinística e intercepto:

Null Hypothesis: INCENTIVO  CONTRABANDO has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

| | | t-Statistic | Prob.* |
|---|---|---|---|
| Augmented Dickey-Fuller test statistic | | -3.941365 | 0.0184 |
| Test critical values: | 1% level | -4.180911 | |
| | 5% level | -3.515523 | |
| | 10% level | -3.188259 | |

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(INCENTIVO  CONTRABANDO)
Method: Least Squares
Date: 10/14/26  Time: 19:10
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments

| Variable | Coefficient | Std. Error | t-Statistic | Prob. |
|---|---|---|---|---|
| INCENTIVO  CONTRABANDO(-1) | -0.553339 | 0.140393 | -3.941365 | 0.0003 |
| C | 0.332058 | 0.123311 | 2.692847 | 0.0102 |
| @TREND("1980") | 0.007811 | 0.004657 | 1.677212 | 0.1011 |

| | | | |
|---|---|---|---|
| R-squared | 0.277855 | Mean dependent var | 0.015094 |
| Adjusted R-squared | 0.242628 | S.D. dependent var | 0.381944 |
| S.E. of regression | 0.332394 | Akaike info criterion | 0.700757 |
| Sum squared resid | 4.529927 | Schwarz criterion | 0.822406 |
| Log likelihood | -12.41665 | Hannan-Quinn criter. | 0.745870 |
| F-statistic | 7.887653 | Durbin-Watson stat | 2.020398 |
| Prob(F-statistic) | 0.001264 | | |

Como la tendencia determinística no resulta significativa (p-valor = 0.1011), vuelvo a hacer el test con intercepto.

Null Hypothesis: INCENTIVO  CONTRABANDO has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

| | | t-Statistic | Prob.* |
|---|---|---|---|
| Augmented Dickey-Fuller test statistic | | -3.525035 | 0.0118 |
| Test critical values: | 1% level | -3.588509 | |
| | 5% level | -2.929734 | |
| | 10% level | -2.603064 | |

*MacKinnon (1996) one-sided p-values.

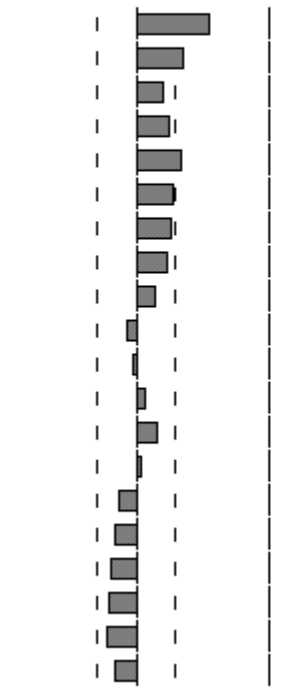
Augmented Dickey-Fuller Test Equation
Dependent Variable: D(INCENTIVO  CONTRABANDO)
Method: Least Squares
Date: 10/14/26  Time: 19:10
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments

| Variable | Coefficient | Std. Error | t-Statistic | Prob. |
|---|---|---|---|---|
| INCENTIVO  CONTRABANDO(-1) | -0.428289 | 0.121499 | -3.525035 | 0.0010 |
| C | 0.396459 | 0.119683 | 3.312581 | 0.0019 |

| | | | |
|---|---|---|---|
| R-squared | 0.228308 | Mean dependent var | 0.015094 |
| Adjusted R-squared | 0.209935 | S.D. dependent var | 0.381944 |
| S.E. of regression | 0.339493 | Akaike info criterion | 0.721662 |
| Sum squared resid | 4.840729 | Schwarz criterion | 0.802761 |
| Log likelihood | -13.87656 | Hannan-Quinn criter. | 0.751737 |
| F-statistic | 12.42587 | Durbin-Watson stat | 2.150524 |
| Prob(F-statistic) | 0.001038 | | |

El intercepto resulta significativo por lo que analizo el p-valor asociado al estadístico tau, como 0.0118<0.05 rechazo la hipótesis nula de raíz unitaria, por lo que proceso a analizar el correlograma de la serie en niveles:

Date: 10/14/26 Time: 19:11
Sample: 1980 2024
Included observations: 45



| Autocorrelation | Partial Correlation | | AC | PAC | Q-Stat | Prob |
|---|---|---|---|---|---|---|
| | | 1 | 0.572 | 0.572 | 15.706 | 0.000 |
| | | 2 | 0.373 | 0.069 | 22.546 | 0.000 |
| | | 3 | 0.211 | -0.039 | 24.795 | 0.000 |
| | | 4 | 0.266 | 0.212 | 28.439 | 0.000 |
| | | 5 | 0.350 | 0.198 | 34.927 | 0.000 |
| | | 6 | 0.289 | -0.049 | 39.459 | 0.000 |
| | | 7 | 0.274 | 0.095 | 43.653 | 0.000 |
| | | 8 | 0.251 | 0.088 | 47.244 | 0.000 |
| | | 9 | 0.144 | -0.162 | 48.462 | 0.000 |
| | | 10 | -0.075 | -0.331 | 48.806 | 0.000 |
| | | 11 | -0.025 | 0.169 | 48.846 | 0.000 |
| | | 12 | 0.071 | 0.109 | 49.166 | 0.000 |
| | | 13 | 0.156 | -0.065 | 50.780 | 0.000 |
| | | 14 | 0.048 | -0.097 | 50.935 | 0.000 |
| | | 15 | -0.126 | -0.061 | 52.054 | 0.000 |
| | | 16 | -0.166 | -0.071 | 54.060 | 0.000 |
| | | 17 | -0.190 | -0.131 | 56.798 | 0.000 |
| | | 18 | -0.218 | -0.090 | 60.527 | 0.000 |
| | | 19 | -0.226 | -0.021 | 64.693 | 0.000 |
| | | 20 | -0.171 | -0.035 | 67.165 | 0.000 |

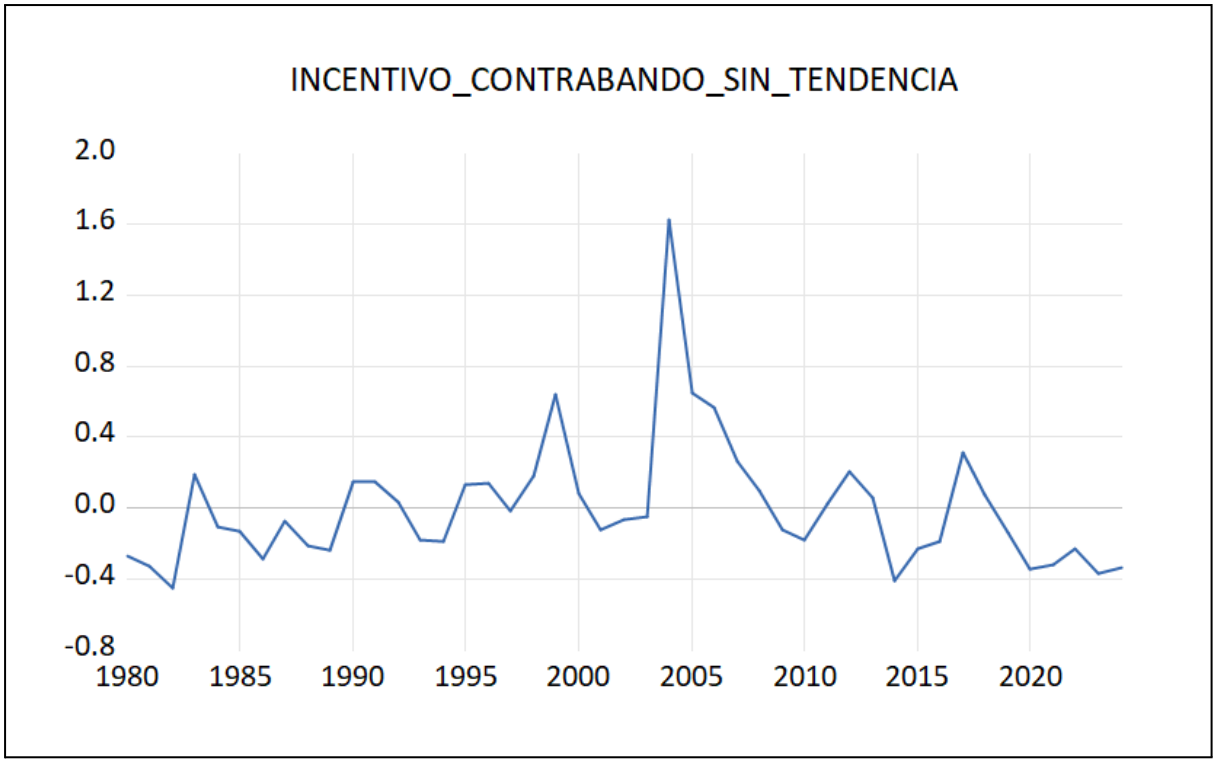
Dependent Variable: INCENTIVO CONTRABANDO
Method: Least Squares
Date: 10/14/26 Time: 19:12
Sample: 1980 2024
Included observations: 45

| Variable | Coefficient | Std. Error | t-Statistic | Prob. |
|---|---|---|---|---|
| @TREND | 0.016547 | 0.004190 | 3.949091 | 0.0003 |
| C | 0.527008 | 0.107046 | 4.923183 | 0.0000 |

| | | | |
|---|---|---|---|
| R-squared | 0.266153 | Mean dependent var | 0.891044 |
| Adjusted R-squared | 0.249087 | S.D. dependent var | 0.421261 |
| S.E. of regression | 0.365045 | Akaike info criterion | 0.865833 |
| Sum squared resid | 5.730078 | Schwarz criterion | 0.946129 |
| Log likelihood | -17.48123 | Hannan-Quinn criter. | 0.895766 |
| F-statistic | 15.59532 | Durbin-Watson stat | 1.094744 |
| Prob(F-statistic) | 0.000286 | | |

**Notar que en el ADF la tendencia no resultaba significativa, sin embargo al estimarla junto al intercepto tiene un p-valor = 0.0003.**

Como tanto la tendencia como el intercepto resultan significativos, hago la serie de los residuos para evaluar la serie sin tendencia:




Date: 10/14/26 Time: 19:23
Sample: 1980 2024
Included observations: 45

| Autocorrelation | Partial Correlation | | AC | PAC | Q-Stat | Prob |
|---|---|---|---|---|---|---|
| | | 1 | 0.436 | 0.436 | 9.1448 | 0.002 |
| | | 2 | 0.186 | -0.005 | 10.854 | 0.004 |
| | | 3 | 0.005 | -0.093 | 10.855 | 0.013 |
| | | 4 | 0.070 | 0.126 | 11.105 | 0.025 |
| | | 5 | 0.192 | 0.168 | 13.048 | 0.023 |
| | | 6 | 0.130 | -0.050 | 13.958 | 0.030 |
| | | 7 | 0.155 | 0.110 | 15.289 | 0.032 |
| | | 8 | 0.170 | 0.123 | 16.937 | 0.031 |
| | | 9 | 0.049 | -0.130 | 17.077 | 0.048 |
| | | 10 | -0.227 | -0.338 | 20.199 | 0.027 |
| | | 11 | -0.174 | 0.093 | 22.082 | 0.024 |
| | | 12 | -0.030 | 0.094 | 22.140 | 0.036 |
| | | 13 | 0.117 | -0.015 | 23.048 | 0.041 |
| | | 14 | 0.002 | -0.128 | 23.048 | 0.059 |
| | | 15 | -0.215 | -0.136 | 26.320 | 0.035 |
| | | 16 | -0.269 | -0.114 | 31.598 | 0.011 |
| | | 17 | -0.288 | -0.144 | 37.884 | 0.003 |
| | | 18 | -0.293 | -0.122 | 44.591 | 0.000 |
| | | 19 | -0.263 | -0.057 | 50.210 | 0.000 |
| | | 20 | -0.171 | -0.085 | 52.686 | 0.000 |

Analizando el correlograma, tanto la FAC como la FACP resultan significativas solamente en el primer rezago (y el rezago 10 de la FACP)

Dependent Variable: INCENTIVO CONTRABANDO SIN TENDENCIA
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/05/26 Time: 11:50
Sample: 1980 2024
Included observations: 45
Convergence achieved after 33 iterations
Coefficient covariance computed using outer product of gradients

| Variable | Coefficient | Std. Error | t-Statistic | Prob. |
|---|---|---|---|---|
| AR(1) | 0.451557 | 0.442462 | 1.020556 | 0.3133 |
| MA(1) | -0.013072 | 0.608852 | -0.021469 | 0.9830 |
| SIGMASQ | 0.102293 | 0.008015 | 12.76265 | 0.0000 |

| | | | |
|---|---|---|---|
| R-squared | 0.196661 | Mean dependent var | 1.78E-16 |
| Adjusted R-squared | 0.158407 | S.D. dependent var | 0.360873 |
| S.E. of regression | 0.331059 | Akaike info criterion | 0.696107 |
| Sum squared resid | 4.603194 | Schwarz criterion | 0.816551 |
| Log likelihood | -12.66240 | Hannan-Quinn criter. | 0.741007 |
| Durbin-Watson stat | 1.971198 | | |

| | |
|---|---|
| Inverted AR Roots | .45 |
| Inverted MA Roots | .01 |

Dependent Variable: INCENTIVO CONTRABANDO SIN TENDENCIA
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/05/26 Time: 14:50
Sample: 1980 2024
Included observations: 45
Convergence achieved after 7 iterations
Coefficient covariance computed using outer product of gradients

| Variable | Coefficient | Std. Error | t-Statistic | Prob. |
|---|---|---|---|---|
| AR(1) | 0.440912 | 0.265595 | 1.660092 | 0.1042 |
| SIGMASQ | 0.102298 | 0.007957 | 12.85634 | 0.0000 |

| | | | |
|---|---|---|---|
| R-squared | 0.196622 | Mean dependent var | 1.78E-16 |
| Adjusted R-squared | 0.177938 | S.D. dependent var | 0.360873 |
| S.E. of regression | 0.327195 | Akaike info criterion | 0.651707 |
| Sum squared resid | 4.603421 | Schwarz criterion | 0.732003 |
| Log likelihood | -12.66341 | Hannan-Quinn criter. | 0.681641 |
| Durbin-Watson stat | 1.976939 | | |

| | |
|---|---|
| Inverted AR Roots | .44 |

Date: 11/05/26 Time: 14:51
Sample: 1980 2024
Q-statistic probabilities adjusted for 1 ARMA term

| Autocorrelation | Partial Correlation | | AC | PAC | Q-Stat | Prob |
|---|---|---|---|---|---|---|
| | | 1 | 0.002 | 0.002 | 0.0002 | |
| | | 2 | 0.043 | 0.043 | 0.0891 | 0.765 |
| | | 3 | -0.128 | -0.128 | 0.9103 | 0.634 |
| | | 4 | 0.001 | -0.000 | 0.9103 | 0.823 |
| | | 5 | 0.185 | 0.200 | 2.7234 | 0.605 |
| | | 6 | 0.007 | -0.012 | 2.7265 | 0.742 |
| | | 7 | 0.069 | 0.051 | 2.9936 | 0.810 |
| | | 8 | 0.128 | 0.189 | 3.9290 | 0.788 |
| | | 9 | 0.107 | 0.107 | 4.6009 | 0.799 |
| | | 10 | -0.265 | -0.337 | 8.8440 | 0.452 |
| | | 11 | -0.102 | -0.083 | 9.4880 | 0.486 |
| | | 12 | -0.011 | 0.064 | 9.4960 | 0.576 |
| | | 13 | 0.183 | 0.060 | 11.718 | 0.469 |
| | | 14 | 0.053 | -0.045 | 11.913 | 0.535 |
| | | 15 | -0.167 | -0.086 | 13.877 | 0.459 |
| | | 16 | -0.116 | -0.085 | 14.862 | 0.461 |
| | | 17 | -0.119 | -0.134 | 15.939 | 0.457 |
| | | 18 | -0.130 | -0.159 | 17.270 | 0.436 |
| | | 19 | -0.141 | -0.090 | 18.884 | 0.399 |
| | | 20 | -0.127 | -0.197 | 20.249 | 0.380 |

Dependent Variable: INCENTIVO CONTRABANDO SIN TENDENCIA
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/05/26 Time: 14:52
Sample: 1980 2024
Included observations: 45
Convergence achieved after 9 iterations
Coefficient covariance computed using outer product of gradients

| Variable | Coefficient | Std. Error | t-Statistic | Prob. |
|---|---|---|---|---|
| AR(1) | 0.438785 | 0.242707 | 1.807877 | 0.0778 |
| AR(10) | -0.239242 | 0.339483 | -0.704724 | 0.4849 |
| SIGMASQ | 0.093367 | 0.010379 | 8.995690 | 0.0000 |

| | | | |
|---|---|---|---|
| R-squared | 0.266762 | Mean dependent var | 1.78E-16 |
| Adjusted R-squared | 0.231846 | S.D. dependent var | 0.360873 |
| S.E. of regression | 0.316285 | Akaike info criterion | 0.621812 |
| Sum squared resid | 4.201511 | Schwarz criterion | 0.742256 |
| Log likelihood | -10.99077 | Hannan-Quinn criter. | 0.666713 |
| Durbin-Watson stat | 1.971948 | | |

| | | | | |
|---|---|---|---|---|
| Inverted AR Roots | .88+.26i | .88-.26i | .56-.69i | .56+.69i |
| | .04-.86i | .04+.86i | -.47+.70i | -.47-.70i |
| | -.79-.27i | -.79+.27i | | |

Dependent Variable: INCENTIVO CONTRABANDO SIN TENDENCIA
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/05/26 Time: 14:51
Sample: 1980 2024
Included observations: 45
Convergence achieved after 6 iterations
Coefficient covariance computed using outer product of gradients

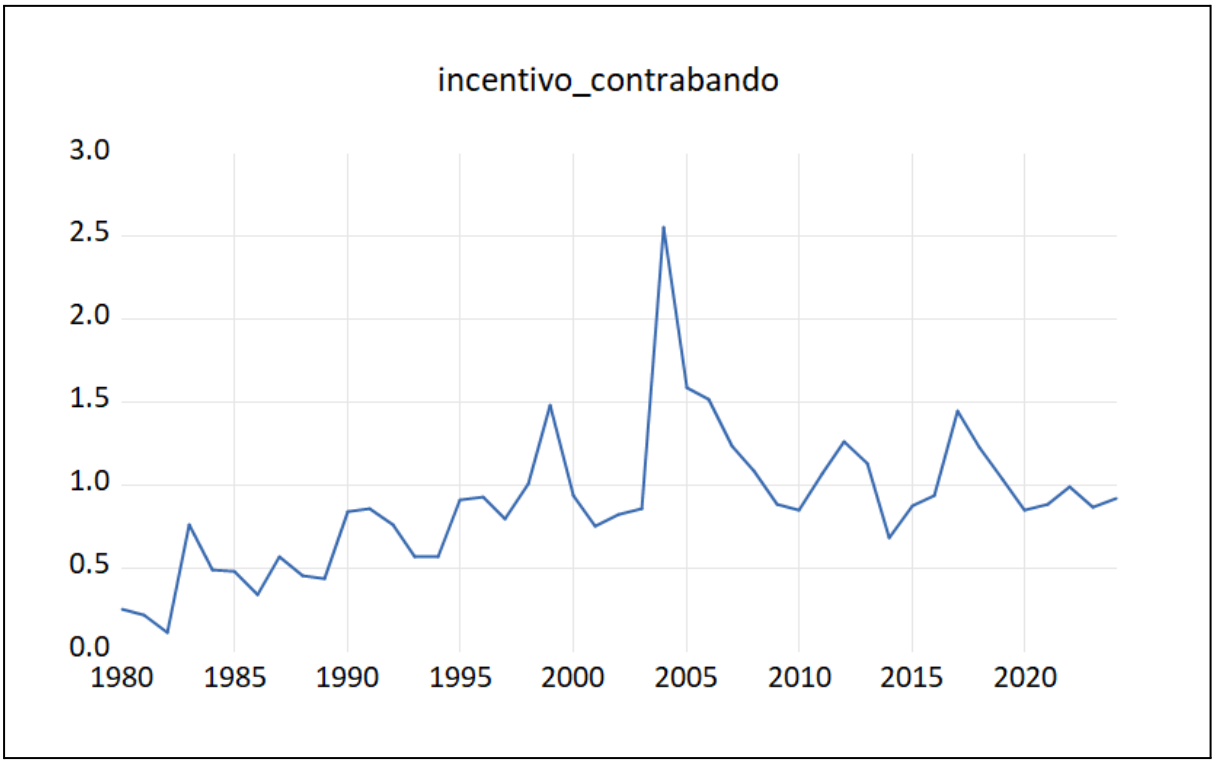
| Variable | Coefficient | Std. Error | t-Statistic | Prob. |
|---|---|---|---|---|
| MA(1) | 0.359727 | 0.267820 | 1.343167 | 0.1863 |
| SIGMASQ | 0.107085 | 0.008679 | 12.33834 | 0.0000 |

| | | | |
|---|---|---|---|
| R-squared | 0.159029 | Mean dependent var | 1.78E-16 |
| Adjusted R-squared | 0.139471 | S.D. dependent var | 0.360873 |
| S.E. of regression | 0.334762 | Akaike info criterion | 0.695714 |
| Sum squared resid | 4.818830 | Schwarz criterion | 0.776010 |
| Log likelihood | -13.65357 | Hannan-Quinn criter. | 0.725648 |
| Durbin-Watson stat | 1.809635 | | |

| | |
|---|---|
| Inverted MA Roots | -.36 |

La estimación del modelo resulta muy difícil, ya que ningún coeficiente resulta significativo, lo cual puede deberse a la presencia de el valor atípico en el año 2004.
Por lo tanto, procedo a incorporar una variable de intervención para solucionar este problema.

Incorporando una Variable de intervención:



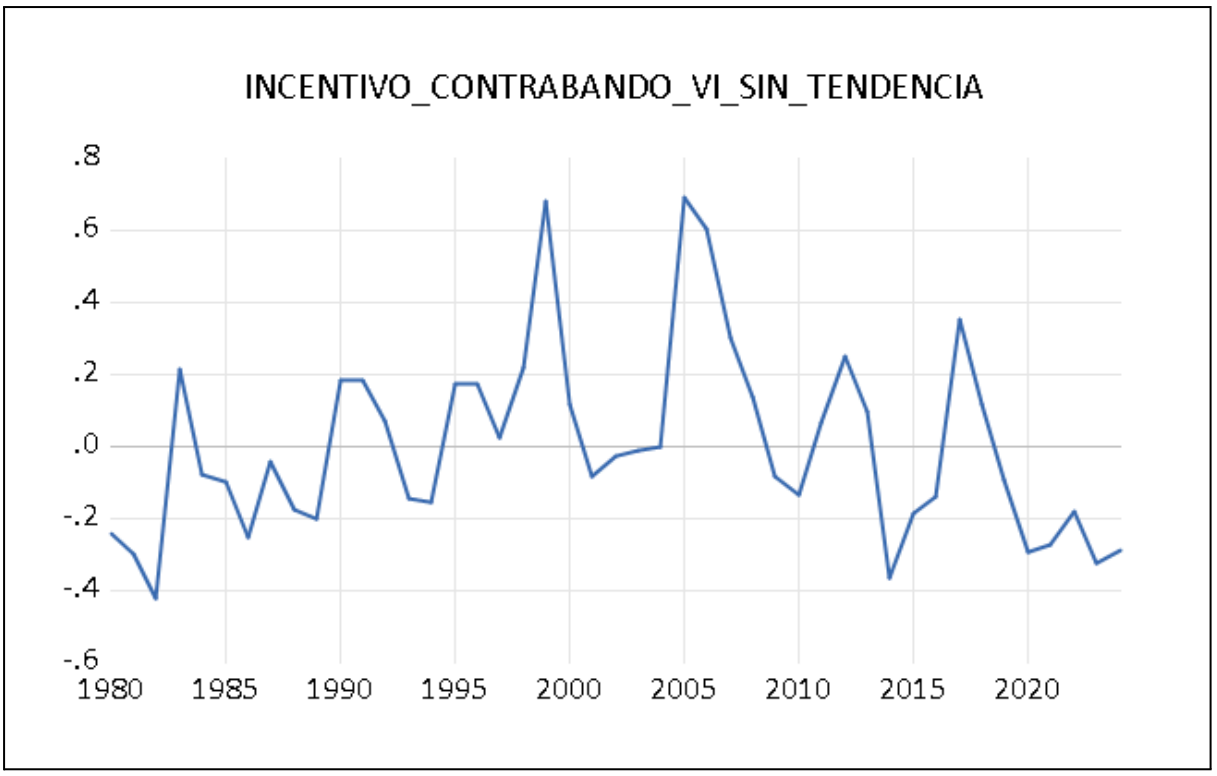

En este caso la variable de intervención se incorpora en el año 2004, debido a que fue el pico de la guerra civil de costa de marfil, que duró desde 2002 hasta 2007.
En ese año el precio de productor en costa de marfil fue extremadamente bajo, comparado con el de Ghana, lo que provoca que el incentivo de contrabando sea tan elevado.

Dependent Variable: INCENTIVO_CONTRABANDO
Method: Least Squares
Date: 11/02/26 Time: 13:16
Sample: 1980 2024
Included observations: 45

| Variable | Coefficient | Std. Error | t-Statistic | Prob. |
|---|---|---|---|---|
| C | 0.499616 | 0.078658 | 6.351742 | 0.0000 |
| @TREND | 0.016108 | 0.003075 | 5.238586 | 0.0000 |
| VI_2004 | 1.667653 | 0.270907 | 6.155822 | 0.0000 |

| | | | |
|---|---|---|---|
| R-squared | 0.614220 | Mean dependent var | 0.891044 |
| Adjusted R-squared | 0.595849 | S.D. dependent var | 0.421261 |
| S.E. of regression | 0.267808 | Akaike info criterion | 0.267244 |
| Sum squared resid | 3.012277 | Schwarz criterion | 0.387688 |
| Log likelihood | -3.012993 | Hannan-Quinn criter. | 0.312144 |
| F-statistic | 33.43515 | Durbin-Watson stat | 0.984937 |
| Prob(F-statistic) | 0.000000 | | |

La variable de impacto incorporada es un impulso aditivo (AO), el cual es una variable que toma valores 0 en todos los años, excepto en 2004, cuando vale 1, se estimó junto a la tendencia y el intercepto y los tres coeficientes resultaron significativos.




Null Hypothesis: INCENTIVO_CONTRABANDO_VI_SIN_TENDENCIA has a unit root
Exogenous: None
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

| | | t-Statistic | Prob.* |
|---|---|---|---|
| Augmented Dickey-Fuller test statistic | | -3.778555 | 0.0003 |
| Test critical values: | 1% level | -2.618579 | |
| | 5% level | -1.948495 | |
| | 10% level | -1.612135 | |

*MacKinnon (1996) one-sided p-values.

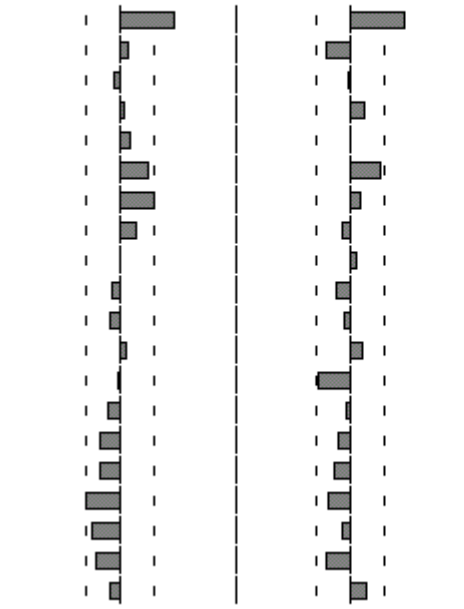
Augmented Dickey-Fuller Test Equation
Dependent Variable: D(INCENTIVO_CONTRABANDO_VI_SIN_TENDENCIA
)
Method: Least Squares
Date: 11/02/26 Time: 13:18
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments

| Variable | Coefficient | Std. Error | t-Statistic | Prob. |
|---|---|---|---|---|
| INCENTIVO_CONTRABANDO_VI_SIN_TE | -0.502591 | 0.133011 | -3.778555 | 0.0005 |

| | | | |
|---|---|---|---|
| R-squared | 0.249257 | Mean dependent var | -0.001013 |
| Adjusted R-squared | 0.249257 | S.D. dependent var | 0.262672 |
| S.E. of regression | 0.227593 | Akaike info criterion | -0.100047 |
| Sum squared resid | 2.227348 | Schwarz criterion | -0.059497 |
| Log likelihood | 3.201025 | Hannan-Quinn criter. | -0.085009 |
| Durbin-Watson stat | 1.814054 | | |

El nuevo ADF sin el impacto de 2004 vuelve a dar una serie estacionaria en niveles.

Date: 11/02/26 Time: 13:18
Sample: 1980 2024
Included observations: 45



| Autocorrelation | Partial Correlation | | AC | PAC | Q-Stat | Prob |
|---|---|---|---|---|---|---|
| | | 1 | 0.483 | 0.483 | 11.235 | 0.001 |
| | | 2 | 0.074 | -0.209 | 11.502 | 0.003 |
| | | 3 | -0.066 | -0.010 | 11.719 | 0.008 |
| | | 4 | 0.038 | 0.131 | 11.793 | 0.019 |
| | | 5 | 0.088 | -0.003 | 12.203 | 0.032 |
| | | 6 | 0.256 | 0.274 | 15.758 | 0.015 |
| | | 7 | 0.302 | 0.097 | 20.824 | 0.004 |
| | | 8 | 0.141 | -0.075 | 21.960 | 0.005 |
| | | 9 | 0.002 | 0.041 | 21.960 | 0.009 |
| | | 10 | -0.081 | -0.116 | 22.358 | 0.013 |
| | | 11 | -0.082 | -0.052 | 22.778 | 0.019 |
| | | 12 | 0.047 | 0.104 | 22.921 | 0.028 |
| | | 13 | -0.015 | -0.281 | 22.935 | 0.042 |
| | | 14 | -0.102 | -0.044 | 23.649 | 0.050 |
| | | 15 | -0.181 | -0.103 | 25.955 | 0.038 |
| | | 16 | -0.185 | -0.153 | 28.448 | 0.028 |
| | | 17 | -0.312 | -0.202 | 35.784 | 0.005 |
| | | 18 | -0.258 | -0.075 | 40.984 | 0.002 |
| | | 19 | -0.219 | -0.208 | 44.876 | 0.001 |
| | | 20 | -0.092 | 0.147 | 45.592 | 0.001 |

Dependent Variable: INCENTIVO CONTRABANDO VI SIN TENDENCIA
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/05/26 Time: 14:58
Sample: 1980 2024
Included observations: 45
Convergence achieved after 6 iterations
Coefficient covariance computed using outer product of gradients

| Variable | Coefficient | Std. Error | t-Statistic | Prob. |
|---|---|---|---|---|
| AR(1) | 0.496312 | 0.138481 | 3.583962 | 0.0009 |
| SIGMASQ | 0.050511 | 0.008554 | 5.904878 | 0.0000 |

| | | | |
|---|---|---|---|
| R-squared | 0.245428 | Mean dependent var | -5.43E-17 |
| Adjusted R-squared | 0.227879 | S.D. dependent var | 0.261650 |
| S.E. of regression | 0.229913 | Akaike info criterion | -0.052520 |
| Sum squared resid | 2.272981 | Schwarz criterion | 0.027776 |
| Log likelihood | 3.181703 | Hannan-Quinn criter. | -0.022587 |
| Durbin-Watson stat | 1.776493 | | |

| | |
|---|---|
| Inverted AR Roots | .50 |

Date: 11/05/26 Time: 14:58
Sample: 1980 2024
Q-statistic probabilities adjusted for 1 ARMA term

| Autocorrelation | Partial Correlation | | AC | PAC | Q-Stat | Prob |
|---|---|---|---|---|---|---|
|  | | 1 | 0.098 | 0.098 | 0.4644 | |
| | | 2 | -0.142 | -0.153 | 1.4494 | 0.229 |
| | | 3 | -0.172 | -0.145 | 2.9332 | 0.231 |
| | | 4 | 0.054 | 0.068 | 3.0826 | 0.379 |
| | | 5 | -0.032 | -0.092 | 3.1353 | 0.535 |
| | | 6 | 0.175 | 0.190 | 4.7875 | 0.442 |
| | | 7 | 0.244 | 0.230 | 8.0987 | 0.231 |
| | | 8 | 0.005 | -0.020 | 8.1003 | 0.324 |
| | | 9 | -0.034 | 0.111 | 8.1674 | 0.417 |
| | | 10 | -0.075 | -0.047 | 8.5092 | 0.484 |
| | | 11 | -0.084 | -0.112 | 8.9504 | 0.537 |
| | | 12 | 0.145 | 0.190 | 10.293 | 0.504 |
| | | 13 | -0.003 | -0.209 | 10.294 | 0.590 |
| | | 14 | -0.049 | -0.067 | 10.455 | 0.656 |
| | | 15 | -0.099 | -0.043 | 11.141 | 0.675 |
| | | 16 | 0.031 | -0.091 | 11.212 | 0.737 |
| | | 17 | -0.222 | -0.196 | 14.949 | 0.528 |
| | | 18 | -0.072 | -0.068 | 15.358 | 0.570 |
| | | 19 | -0.139 | -0.322 | 16.926 | 0.528 |
| | | 20 | -0.021 | -0.025 | 16.964 | 0.592 |

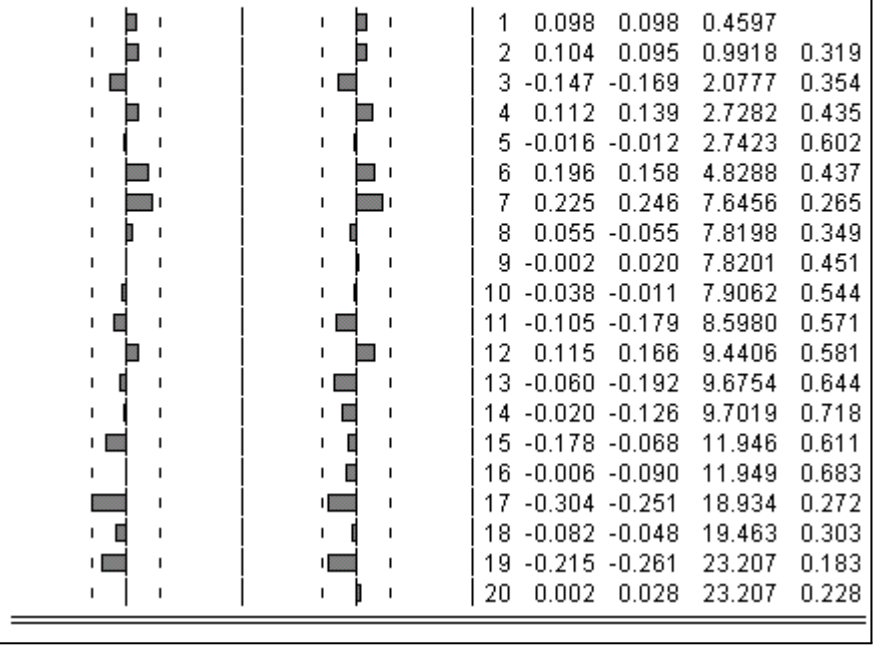
Dependent Variable: INCENTIVO_CONTRABANDO_VI_SIN_TENDENCIA
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/02/26 Time: 13:20
Sample: 1980 2024
Included observations: 45
Convergence achieved after 42 iterations
Coefficient covariance computed using outer product of gradients

| Variable | Coefficient | Std. Error | t-Statistic | Prob. |
|---|---|---|---|---|
| MA(1) | 0.478994 | 0.150264 | 3.187685 | 0.0027 |
| SIGMASQ | 0.050568 | 0.008774 | 5.763403 | 0.0000 |

| | | | |
|---|---|---|---|
| R-squared | 0.244574 | Mean dependent var | -5.43E-17 |
| Adjusted R-squared | 0.227006 | S.D. dependent var | 0.261650 |
| S.E. of regression | 0.230043 | Akaike info criterion | -0.051882 |
| Sum squared resid | 2.275551 | Schwarz criterion | 0.028414 |
| Log likelihood | 3.167355 | Hannan-Quinn criter. | -0.021949 |
| Durbin-Watson stat | 1.771654 | | |

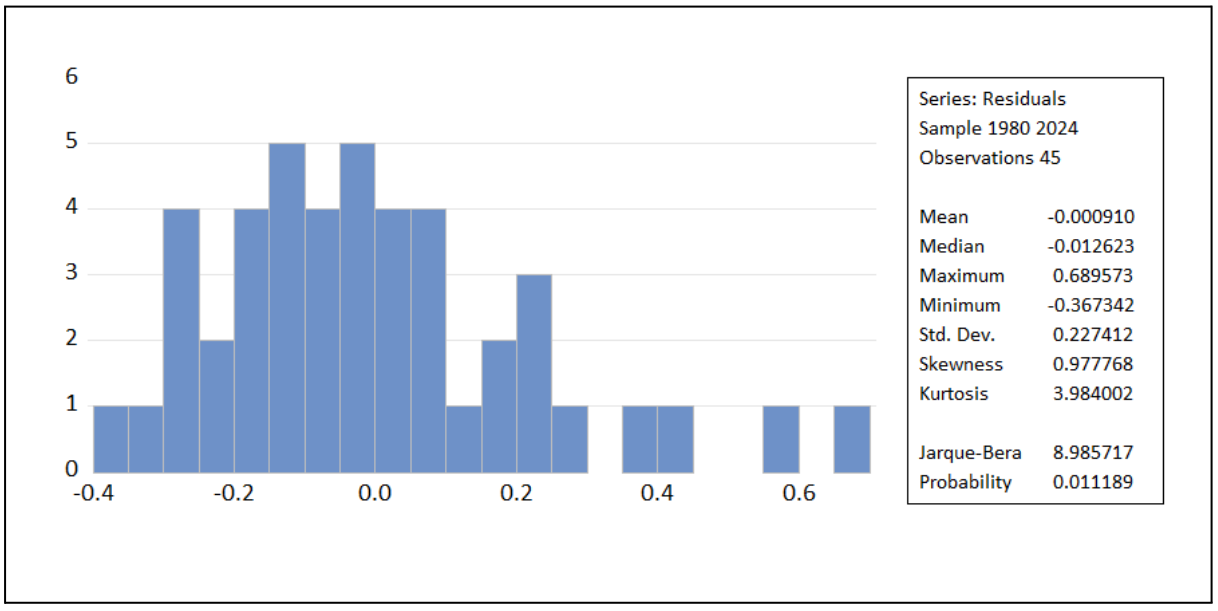
| | |
|---|---|
| Inverted MA Roots | -.48 |

Date: 11/02/26 Time: 13:21
Sample: 1980 2024
Q-statistic probabilities adjusted for 1 ARMA term



| Autocorrelation | Partial Correlation | | AC | PAC | Q-Stat | Prob |
|---|---|---|---|---|---|---|
| | | 1 | 0.098 | 0.098 | 0.4597 | |
| | | 2 | 0.104 | 0.095 | 0.9918 | 0.319 |
| | | 3 | -0.147 | -0.169 | 2.0777 | 0.354 |
| | | 4 | 0.112 | 0.139 | 2.7282 | 0.435 |
| | | 5 | -0.016 | -0.012 | 2.7423 | 0.602 |
| | | 6 | 0.196 | 0.158 | 4.8288 | 0.437 |
| | | 7 | 0.225 | 0.246 | 7.6456 | 0.265 |
| | | 8 | 0.055 | -0.055 | 7.8198 | 0.349 |
| | | 9 | -0.002 | 0.020 | 7.8201 | 0.451 |
| | | 10 | -0.038 | -0.011 | 7.9062 | 0.544 |
| | | 11 | -0.105 | -0.179 | 8.5980 | 0.571 |
| | | 12 | 0.115 | 0.166 | 9.4406 | 0.581 |
| | | 13 | -0.060 | -0.192 | 9.6754 | 0.644 |
| | | 14 | -0.020 | -0.126 | 9.7019 | 0.718 |
| | | 15 | -0.178 | -0.068 | 11.946 | 0.611 |
| | | 16 | -0.006 | -0.090 | 11.949 | 0.683 |
| | | 17 | -0.304 | -0.251 | 18.934 | 0.272 |
| | | 18 | -0.082 | -0.048 | 19.463 | 0.303 |
| | | 19 | -0.215 | -0.261 | 23.207 | 0.183 |
| | | 20 | 0.002 | 0.028 | 23.207 | 0.228 |

Comparo ambos modelos según el criterio de Akaike, en donde gana el modelo con AR(1) (-0.05252 < -0.05188), pero como la diferencia es tan pequeña, de tan solo, 0.00064, elijo el modelo MA(1) por presentar un mejor test de Jarque-Bera.




Al agregar esta primera variable de intervención se ve que se pudo obtener un modelo con residuos ruido blanco y coeficientes significativos, sin embargo sigue siendo problemático el hecho de que Dickey Fuller rechace la presencia de tendencia determinística y sin embargo resulta muy significativa en las estimaciones, a pesar de que analizando el gráfico solo se ve una tendencia en el periodo 1983-2000, lo que concuerda con las reformas económicas

de Ghana (ERP) ocurrida durante ese periodo, por lo tanto incorporo una nueva variable de intervención que refleje ese cambio de pendiente (SLS), la cual esta construida con valores igual a 0 para el periodo anterior a 1983, y luego tiene un crecimiento lineal (1, 2, 3, …) desde 1983 hasta el año 2000, y valores iguales a los de el año 2000 (18) hasta finalizar la serie.
Además la variable de intervención para el año 2004, dejó de ser una variable aditiva de única intervención, sino que pasó a ser de cambio transitorio (TC), que tiene su pico máximo en 2004, y su impacto va decayendo año a año a un nivel gamma (δ) de 0.5.

Dependent Variable: INCENTIVO CONTRABANDO
Method: Least Squares
Date: 11/05/26 Time: 11:55
Sample: 1980 2024
Included observations: 45

| Variable | Coefficient | Std. Error | t-Statistic | Prob. |
|---|---|---|---|---|
| TC 2004 | 1.511684 | 0.179457 | 8.423677 | 0.0000 |
| SLS IC | 0.037937 | 0.009592 | 3.954919 | 0.0003 |
| @TREND | -0.000742 | 0.004598 | -0.161420 | 0.8726 |
| C | 0.332188 | 0.067687 | 4.907727 | 0.0000 |

| | | | |
|---|---|---|---|
| R-squared | 0.807315 | Mean dependent var | 0.891044 |
| Adjusted R-squared | 0.793216 | S.D. dependent var | 0.421261 |
| S.E. of regression | 0.191562 | Akaike info criterion | -0.382520 |
| Sum squared resid | 1.504540 | Schwarz criterion | -0.221928 |
| Log likelihood | 12.60670 | Hannan-Quinn criter. | -0.322653 |
| F-statistic | 57.26068 | Durbin-Watson stat | 1.665702 |
| Prob(F-statistic) | 0.000000 | | |

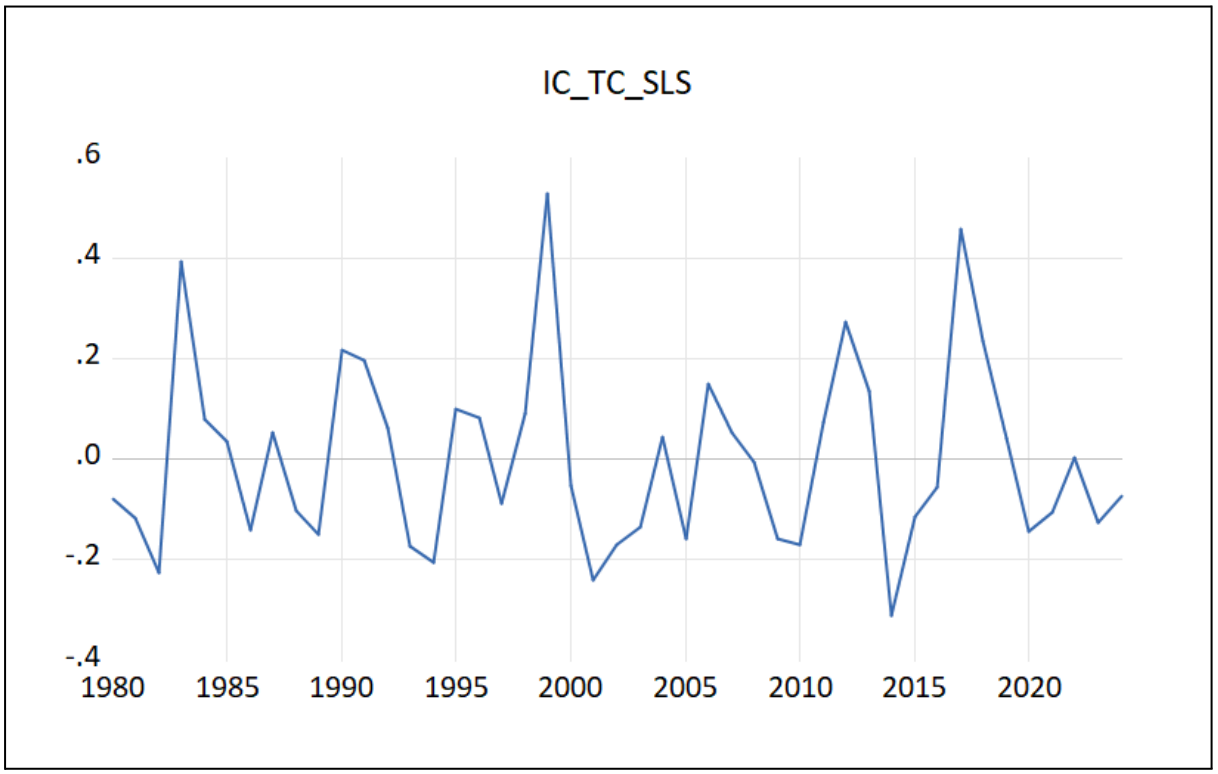
De este modo la tendencia determinística deja de ser significativa, por lo que no se encuentra una tendencia lineal que abarque todo el periodo 1980-2024, ya que la variable de intervención absorbió ese impacto.

Dependent Variable: INCENTIVO_CONTRABANDO
Method: Least Squares
Date: 11/04/26 Time: 22:43
Sample: 1980 2024
Included observations: 45

| Variable | Coefficient | Std. Error | t-Statistic | Prob. |
|---|---|---|---|---|
| TC_2004 | 1.518167 | 0.172864 | 8.782426 | 0.0000 |
| C | 0.333785 | 0.066179 | 5.043689 | 0.0000 |
| SLS_IC | 0.036578 | 0.004542 | 8.053344 | 0.0000 |

| | | | |
|---|---|---|---|
| R-squared | 0.807192 | Mean dependent var | 0.891044 |
| Adjusted R-squared | 0.798011 | S.D. dependent var | 0.421261 |
| S.E. of regression | 0.189328 | Akaike info criterion | -0.426329 |
| Sum squared resid | 1.505497 | Schwarz criterion | -0.305885 |
| Log likelihood | 12.59241 | Hannan-Quinn criter. | -0.381429 |
| F-statistic | 87.91671 | Durbin-Watson stat | 1.662997 |
| Prob(F-statistic) | 0.000000 | | |

A simple vista se ve como tanto la varianza como el AIC mejoró con respecto a los modelos anteriores sin variable de intervención y con una única variable de intervención aditiva.



Null Hypothesis: IC_TC_SLS has a unit root
Exogenous: None
Lag Length: 1 (Automatic - based on SIC, maxlag=9)

| | | t-Statistic | Prob.* |
|---|---|---|---|
| Augmented Dickey-Fuller test statistic | | -6.093932 | 0.0000 |
| Test critical values: | 1% level | -2.619851 | |
| | 5% level | -1.948686 | |
| | 10% level | -1.612036 | |

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(IC_TC_SLS)
Method: Least Squares
Date: 11/04/26 Time: 22:44
Sample (adjusted): 1982 2024
Included observations: 43 after adjustments

| Variable | Coefficient | Std. Error | t-Statistic | Prob. |
|---|---|---|---|---|
| IC_TC_SLS(-1) | -1.147134 | 0.188242 | -6.093932 | 0.0000 |
| D(IC_TC_SLS(-1)) | 0.364987 | 0.145463 | 2.509134 | 0.0161 |

| | | | |
|---|---|---|---|
| R-squared | 0.498541 | Mean dependent var | 0.000975 |
| Adjusted R-squared | 0.486311 | S.D. dependent var | 0.244087 |
| S.E. of regression | 0.174942 | Akaike info criterion | -0.603329 |
| Sum squared resid | 1.254794 | Schwarz criterion | -0.521412 |
| Log likelihood | 14.97157 | Hannan-Quinn criter. | -0.573121 |
| Durbin-Watson stat | 2.236215 | | |

Date: 11/04/26 Time: 22:44
Sample: 1980 2024
Included observations: 45

| Autocorrelation | Partial Correlation | | AC | PAC | Q-Stat | Prob |
|---|---|---|---|---|---|---|
| | | 1 | 0.165 | 0.165 | 1.3011 | 0.254 |
| | | 2 | -0.325 | -0.362 | 6.5086 | 0.039 |
| | | 3 | -0.407 | -0.324 | 14.838 | 0.002 |
| | | 4 | -0.034 | -0.041 | 14.896 | 0.005 |
| | | 5 | 0.065 | -0.200 | 15.123 | 0.010 |
| | | 6 | -0.035 | -0.251 | 15.188 | 0.019 |
| | | 7 | 0.144 | 0.132 | 16.339 | 0.022 |
| | | 8 | 0.121 | -0.045 | 17.175 | 0.028 |
| | | 9 | 0.035 | 0.002 | 17.248 | 0.045 |
| | | 10 | -0.281 | -0.201 | 22.023 | 0.015 |
| | | 11 | -0.136 | -0.046 | 23.174 | 0.017 |
| | | 12 | 0.073 | -0.061 | 23.516 | 0.024 |
| | | 13 | 0.129 | -0.101 | 24.620 | 0.026 |
| | | 14 | 0.033 | -0.069 | 24.695 | 0.038 |
| | | 15 | -0.076 | -0.117 | 25.105 | 0.049 |
| | | 16 | 0.067 | 0.006 | 25.433 | 0.063 |
| | | 17 | -0.138 | -0.281 | 26.863 | 0.060 |
| | | 18 | -0.039 | -0.032 | 26.985 | 0.079 |
| | | 19 | 0.011 | -0.091 | 26.995 | 0.105 |
| | | 20 | 0.060 | -0.271 | 27.294 | 0.127 |

Dependent Variable: IC_TC_SLS
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/04/26 Time: 22:52
Sample: 1980 2024
Included observations: 45
Convergence achieved after 12 iterations
Coefficient covariance computed using outer product of gradients

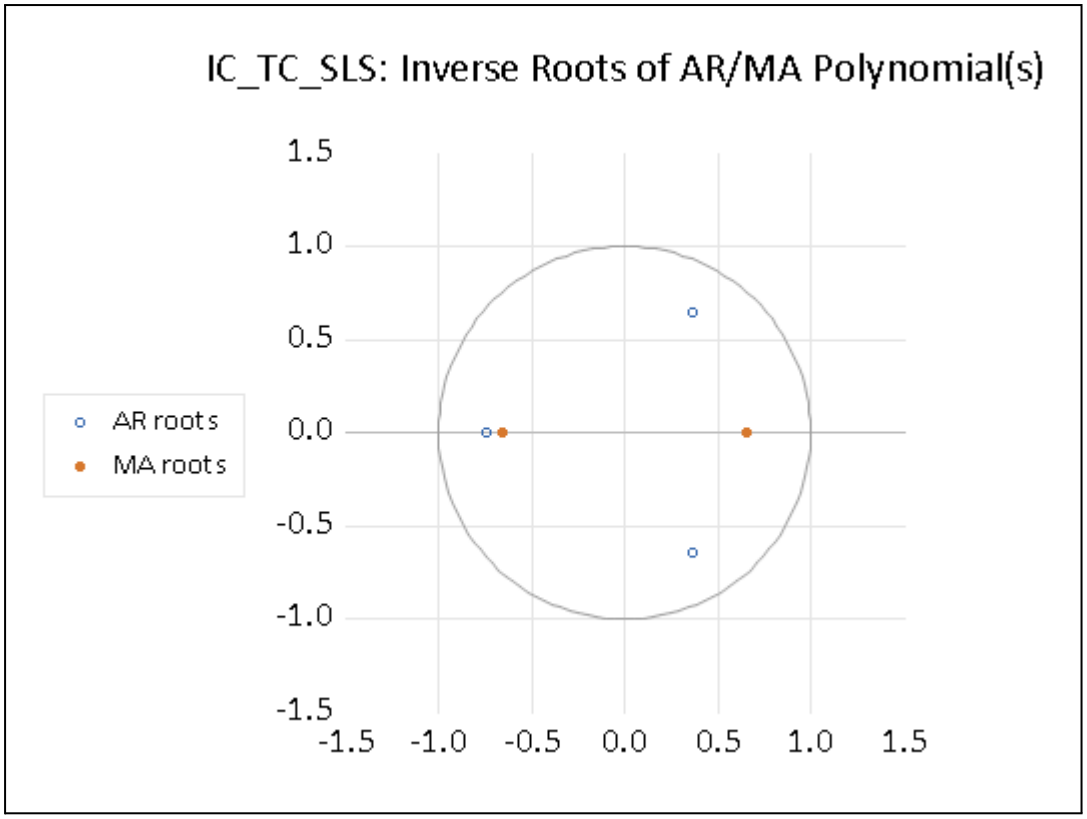
| Variable | Coefficient | Std. Error | t-Statistic | Prob. |
|---|---|---|---|---|
| AR(3) | -0.406601 | 0.189518 | -2.145447 | 0.0377 |
| MA(2) | -0.423696 | 0.205844 | -2.058337 | 0.0458 |
| SIGMASQ | 0.023659 | 0.003976 | 5.950293 | 0.0000 |

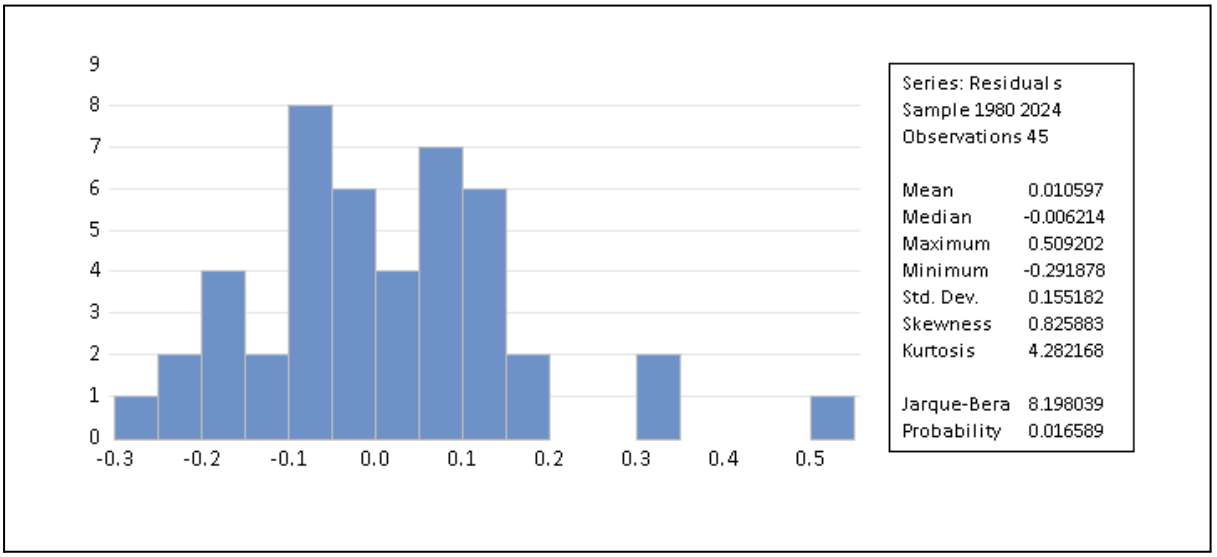
| | | | |
|---|---|---|---|
| R-squared | 0.292832 | Mean dependent var | -8.64E-17 |
| Adjusted R-squared | 0.259158 | S.D. dependent var | 0.184975 |
| S.E. of regression | 0.159212 | Akaike info criterion | -0.752538 |
| Sum squared resid | 1.064638 | Schwarz criterion | -0.632094 |
| Log likelihood | 19.93210 | Hannan-Quinn criter. | -0.707637 |
| Durbin-Watson stat | 1.962562 | | |

| | | | |
|---|---|---|---|
| Inverted AR Roots | .37-.64i | .37+.64i | -.74 |
| Inverted MA Roots | .65 | -.65 | |

Date: 11/04/26 Time: 22:54
Sample: 1980 2024
Q-statistic probabilities adjusted for 2 ARMA terms

| Autocorrelation | Partial Correlation | | AC | PAC | Q-Stat | Prob |
|---|---|---|---|---|---|---|
| | | 1 | 0.006 | 0.006 | 0.0019 | |
| | | 2 | 0.001 | 0.001 | 0.0019 | |
| | | 3 | -0.122 | -0.122 | 0.7455 | 0.388 |
| | | 4 | 0.018 | 0.020 | 0.7625 | 0.683 |
| | | 5 | -0.020 | -0.021 | 0.7842 | 0.853 |
| | | 6 | -0.228 | -0.246 | 3.5978 | 0.463 |
| | | 7 | 0.083 | 0.099 | 3.9854 | 0.552 |
| | | 8 | -0.003 | -0.011 | 3.9859 | 0.679 |
| | | 9 | 0.100 | 0.039 | 4.5679 | 0.713 |
| | | 10 | -0.266 | -0.255 | 8.8574 | 0.354 |
| | | 11 | -0.102 | -0.118 | 9.5017 | 0.392 |
| | | 12 | -0.042 | -0.082 | 9.6146 | 0.475 |
| | | 13 | -0.004 | -0.039 | 9.6157 | 0.565 |
| | | 14 | -0.021 | -0.066 | 9.6465 | 0.647 |
| | | 15 | -0.157 | -0.187 | 11.386 | 0.579 |
| | | 16 | 0.169 | 0.023 | 13.464 | 0.490 |
| | | 17 | -0.175 | -0.267 | 15.787 | 0.396 |
| | | 18 | 0.057 | -0.023 | 16.040 | 0.450 |
| | | 19 | 0.073 | 0.115 | 16.475 | 0.490 |
| | | 20 | 0.023 | -0.167 | 16.520 | 0.556 |




Este modelo luego de aplicar las variables de intervención, resulta un AR(3) MA(2) con ambos coeficientes significativos al 5% utilizando el modelo de estimación de Máxima Verosimilitud (ML), ,con residuos ruido blanco, y raíces invertidas dentro del círculo unitario, y con un criterio de Akaike de -0.75, muy inferior al del modelo anterior con variable de intervención (-0.70 < 0.03).